Numerical instability

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☆ Numerical instability of a numerical method → How to make a numerical method stable?

First, we examine how LTE is propagated.

- Let e_n denote the total error. $e_n = y(x_n) - y_n$, where y_n is numerical solution / approximation to $y(x_n)$ by some numerical method.
- Let us consider the total error at the next step.

$$e_{n+1} = y(x_{n+1}) - y_{n+1}$$
$$= e^{\lambda h} y(x_n) - E(\lambda h) y_n$$

$e_n = y(x_n) - y_n \Rightarrow y_n = y(x_n) - e_n$

$$\therefore e_{n+1} = e^{\lambda h} y(x_n) - E(\lambda h)(y(x_n) - e_n))$$

$$\boxed{e_{n+1} = \underbrace{\left(e^{\lambda h} - E(\lambda h)\right) y(x_n)}_{\text{local truncation error (LTE)}} + \underbrace{E(\lambda h)\, e_n}_{\text{propagation error}}}$$

Again, $e_{n+2} = y(x_{n+2}) - y_{n+2}$

$$= e^{\lambda h} y(x_{n+1}) - E(\lambda h) y_{n+1}$$
$$= e^{\lambda h} \cdot e^{\lambda h} y(x_n) - E(\lambda h) . E(\lambda h) y_n$$
$$= (e^{\lambda h})^2 y(x_n) - E(\lambda h)^2 y_n$$
$$= (e^{\lambda h})^2 y(x_n) - E(\lambda h)^2 (y(x_n) - e_n)$$

$$\boxed{e_{n+2} = \underbrace{\left[(e^{\lambda h})^2 - E(\lambda h)^2\right] y(x_n)}_{\text{LTE}} + \underbrace{E(\lambda h)^2 e_n}_{\substack{\text{propagation error} \\ \text{(propagation of error at } x_n)}}}$$

The total error committed at x_n is magnified by the factor $E(\lambda h)^2$.

$\therefore E(\lambda h)$ is called the growth factor.

At the $(n+k)^{th}$ step, the error at x_n gets magnified by $E(\lambda h)^k$.

$\therefore$ Unless the propagation is controlled, numerical instability may be caused. So, we choose λh s.t. $|E(\lambda h)| < 1$.

$$\boxed{e_{n+k} = \left[(e^{\lambda h})^k - E(\lambda h)^k\right] y(x_n) + E(\lambda h)^k e_n}$$

- Absolute stability → A numerical method (single step) is said to be absolutely stable if $|E(\lambda h)| < 1$
- Relative stability → A numerical method is said to be relatively stable if $|E(\lambda h)| < e^{\lambda h}$
- Interval of absolute stability → An interval (a, b) s.t. whenever $h\lambda \in (a, b)$, the method is absolutely stable, such an interval is called interval of absolute stability.

Intervals of absolute stability of RK methods →

① Euler's method $\quad E(\lambda h) = 1 + \lambda h$

$|E(\lambda h)| < 1 \Rightarrow |1 + \lambda h| < 1$

$\Rightarrow -1 < 1 + \lambda h < 1$

$\Rightarrow -2 < \lambda h < 0$

$\therefore$ For Euler's method, interval of absolute stability is $\lambda h \in (-2, 0)$. Since $h > 0$, we get that $\lambda < 0$.

Now, since $\lambda < 0$, we choose the step size h so small that $h < 2/|\lambda|$. Then, if $|\lambda|$ is very large, we must choose a very small step length h in the Euler's method, to get a decreasing solution.

② 2nd order RK method $\quad E(\lambda h) = 1 + \lambda h + \frac{\lambda^2 h^2}{2}$

$|E(\lambda h)| < 1 \Rightarrow \left|1 + \lambda h + \frac{\lambda^2 h^2}{2}\right| < 1$

$\Rightarrow -1 < 1 + \lambda h + \frac{\lambda^2 h^2}{2} < 1$

$\Rightarrow -2 < \lambda h + \frac{\lambda^2 h^2}{2} < 0$

$\Rightarrow -4 < 2\lambda h + \lambda^2 h^2 < 0$

$\lambda^2 h^2 + 2\lambda h + 4 > 0 \qquad\qquad \lambda^2 h^2 + 2\lambda h < 0$

$(1 + \lambda h)^2 + 3 > 0 \qquad\qquad \lambda h(\lambda h + 2) < 0$

↳ true $\forall\, \lambda h \in \mathbb{R} \qquad\qquad \lambda h \in (-2, 0)$

③ 3rd order RK method $\quad E(\lambda h) = 1 + \lambda h + \frac{\lambda^2 h^2}{2} + \frac{\lambda^3 h^3}{3!}$

$|E(\lambda h)| < 1 \Rightarrow -2.51 < \lambda h < 0$

④ 4th order RK method $\quad E(\lambda h) = 1 + \lambda h + \frac{\lambda^2 h^2}{2!} + \frac{\lambda^3 h^3}{3!} + \frac{\lambda^4 h^4}{4!}$

$|E(\lambda h)| < 1 \Rightarrow -2.785 < \lambda h < 0$

$\therefore h < \frac{2.785}{|\lambda|}$

⑤ Trapezoidal method

$$y_{n+1} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right]$$

Apply to the test eq. $y' = \lambda y$.

$f(x, y) = y' = \lambda y$

$f(x_n, y_n) = \lambda y_n \qquad f(x_{n+1}, y_{n+1}) = \lambda y_{n+1}$

$y_{n+1} = y_n + \frac{h}{2}\left[\lambda y_n + \lambda y_{n+1}\right]$

$= y_n + \frac{\lambda h}{2}\left[y_n + y_{n+1}\right]$

$\left(1 - \frac{\lambda h}{2}\right) y_{n+1} = \left(1 + \frac{\lambda h}{2}\right) y_n$

$$y_{n+1} = \left(\frac{1 + \frac{\lambda h}{2}}{1 - \frac{\lambda h}{2}}\right) y_n$$

It can be seen that $\left|\frac{1 + \frac{\lambda h}{2}}{1 - \frac{\lambda h}{2}}\right| < 1 \quad \forall\, \lambda < 0$ and $h > 0$.

$\therefore$ This method is unconditionally stable (stability is independent of step length).

$y' = \lambda y, \; y(0) = 1$ in the general case where λ is a complex no. with −ve real part.

The exact solution is $y(x) = e^{\lambda x}$.

If $\lambda = -k + iv$ (i denotes imaginary unit, $i^2 = -1$), $k > 0$.

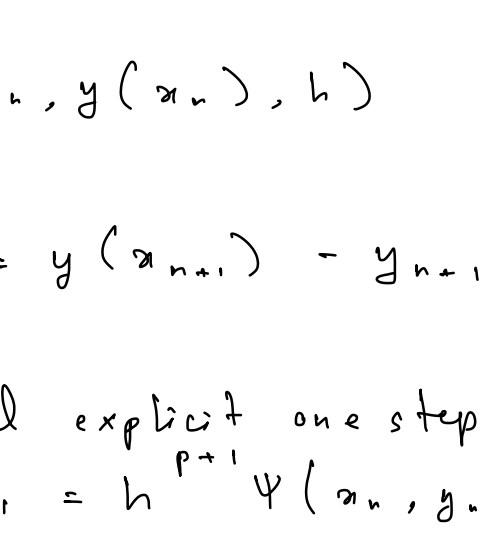
$y(x) = e^{\lambda x} = e^{-kx}(\cos vx + i \sin vx)$

Both real & imaginary part of the equation are damped oscillations, as $|e^{\lambda x}| = e^{\lambda x}$ is a decreasing exponential.

The requirement that numerical method gives a decreasing seq. led to the following condition →

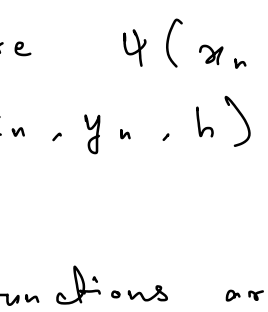
① Euler method → $|1 + \lambda h| < 1$

$|\lambda h - (-1)| < 1$



② 2nd order RK method $\left|1 + \lambda h + \frac{\lambda^2 h^2}{2}\right| < 1$

③ Trapezoidal method



Stability region

($\forall\, \lambda < 0$ and $h > 0$, method is stable)

$$\left|\frac{1 + \frac{\lambda h}{2}}{1 - \frac{\lambda h}{2}}\right| < 1$$

General explicit one-step method

$y_{n+1} = y_n + h\phi(x_n, y_n, h)$, where $\phi(x_n, y_n, h)$ is called increment function. — ①

LTE $\quad T_{n+1} = y(x_{n+1}) - y(x_n) - h\phi(x_n, y(x_n), h)$

Global error / Total truncation error $\quad e_{n+1} = y(x_{n+1}) - y_{n+1}$

Order of a method → If LTE of a general explicit one step method defined by ① is of order p if $T_{n+1} = h^{p+1}\psi(x_n, y_n, h) + O(h^{p+2})$, where $\psi(x_n, y_n, h)$ is called principal error function and $h\psi(x_n, y_n, h)$ is called principal LTE term.

☆☆ A method in which R functions are defined is called R stage RK method.
↳ $k_1, k_2 \dots k_R$

If p is the max. order of a method with R stages, then it is seen that

$p = R$ for $R \le 4$

$p = R - 1$ for $R = 5, 6, 7$

$p = R - 2$ for $R \geq 8$

Stability of ① The method ① is said to be stable if for each diff. eq. $y' = f(x, y)$, $y(x_0) = y_0$ with $f(x, y)$ satisfying Lipschitz condition, ∃ +ve constants h and k ∋ the diff. of numerical solutions obtained from ① satisfies

$$\boxed{|y_n - \tilde{y}_n| \le k|y_0 - \tilde{y}_0| \quad \forall\, 0 \le h \le h_0}$$

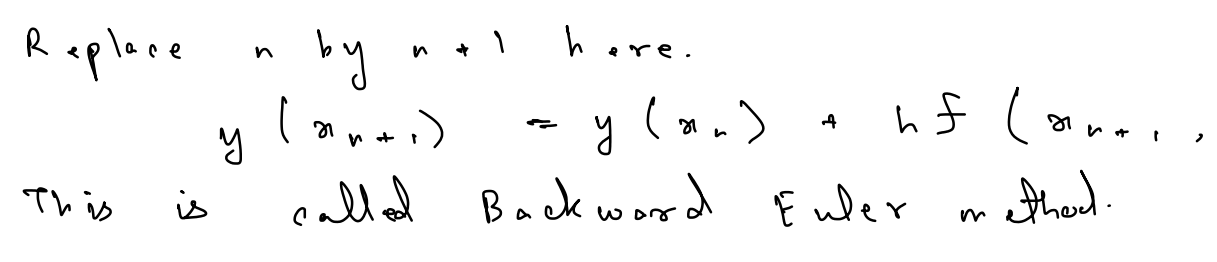
(small changes in the data produces bounded change in the solution)

Consistent ① is consistent if $\phi(x, y, 0) = f(x, y)$

Convergence The method ① is said to be convergent if $y_n \to y(x_n)$ $\forall\, 0 \le x_n \le b$ as $n \to \infty$, $h = b/n$ for any diff. eq. $y' = f(x, y)$, $y(0) = y_0$ with $f(x, y)$ satisfying Lipschitz condition.

Theorems

1. If $\phi(x, y, h)$ (increment function) satisfies Lipschitz condition, then method ① is stable.
2. Let the function $\phi(x, y, h)$ be continuous in the region defined by $x \in [a, b]$, $y \in (-\infty, \infty)$, $0 \le h \le h_0$, where $h_0 > 0$ and there exists a constant L s.t. $|\phi(x, y^*, h) - \phi(x, y, h)| \le L|y^* - y|$ for all (x, y, h) and (x, y^*, h) in the region defined. The relation $\phi(x, y, 0) = f(x, y)$ is a necessary & sufficient condition for convergence of the method defined by the increment function $\phi(x, y, h)$.



Condition is called condition of consistency. Roughly speaking the theorem states that consistency is the necessity & sufficient condition for convergence

$y' = f(x, y), \quad y(x_0) = y_0$

① Forward approx. of derivative →

$$\boxed{y'(x_n) = \frac{y(x_{n+1}) - y(x_n)}{h}}$$

$\frac{y(x_{n+1}) - y(x_n)}{h} \approx f(x_n, y(x_n))$

$y(x_{n+1}) \approx y(x_n) + hf(x_n, y(x_n))$ → This is Euler's method

② Backward diff. approx. of derivative →

$$\boxed{y'(x_n) = \frac{y(x_n) - y(x_{n-1})}{h}}$$

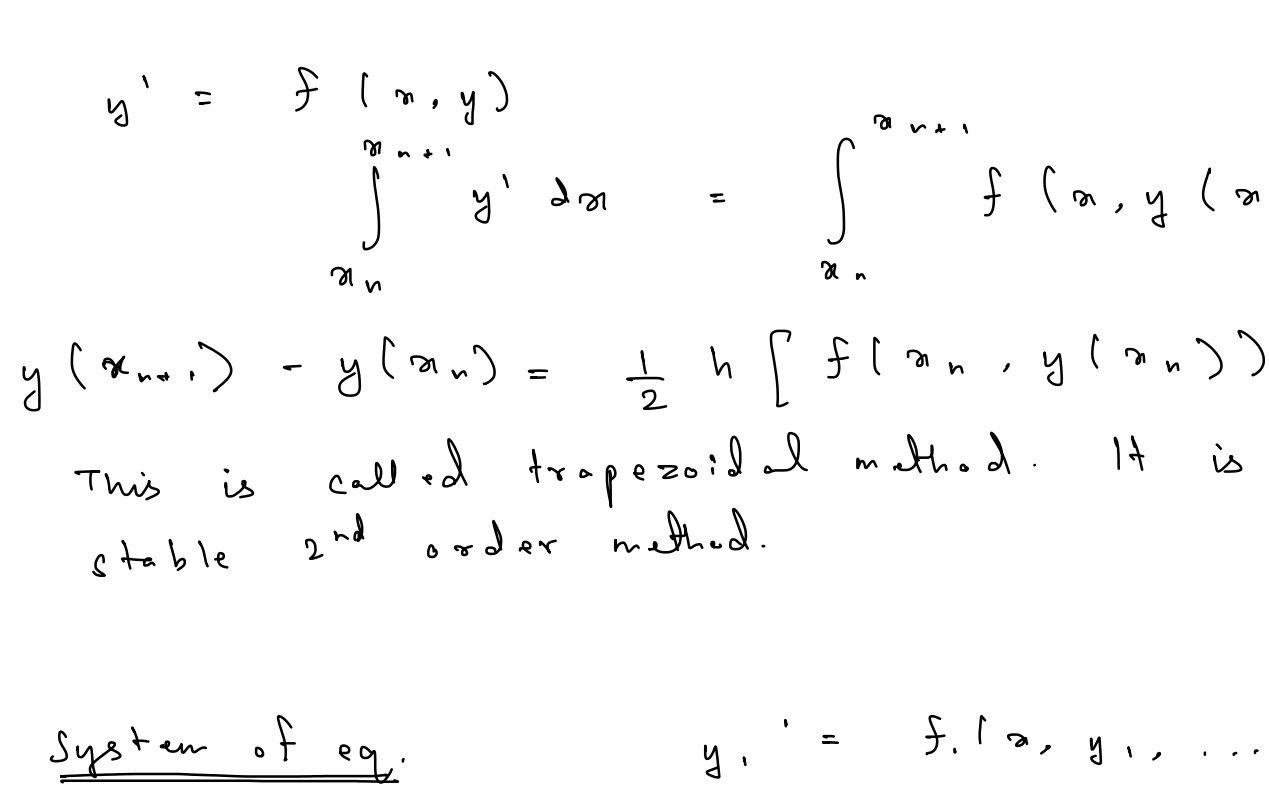
$y(x_n) = y(x_{n-1}) + hy'(x_n)$

$y(x_n) = y(x_{n-1}) + hf(x_n, y(x_n))$

Replace n by $n+1$ here.

$y(x_{n+1}) = y(x_n) + hf(x_{n+1}, y_{n+1})$

This is called Backward Euler method.



$\boxed{y_n = y_{n-1} + hf(x_n, y_n)}$

replace n by n+1

OR

$\boxed{y_{n+1} = y_n + hf(x_n, y_{n+1})}$

Backward Euler's method

↓

This is unconditionally stable.

Implicit method

$y' = \lambda y$, λ is −ve

$y_{n+1} = y_n + h\lambda y_{n+1}$

$(1 - \lambda h) y_{n+1} = y_n$

$y_{n+1} = \left(\frac{1}{1 - \lambda h}\right) y_n$

$\left|\frac{1}{1 - \lambda h}\right| < 1$, λ −ve, $h > 0$

③ Central diff. approx. →

$$y'(x_n) = \frac{y(x_{n+1}) - y(x_{n-1})}{2h}$$

$y(x_{n+1}) = y(x_{n-1}) + 2hy'(x_n)$

$y(x_{n+1}) = y(x_{n-1}) + 2hf(x_n, y(x_n))$

$\therefore y_{n+1} = y_{n-1} + 2hf(x_n, y_n)$ — This is called midpoint method

It is a 2nd order, 2 point, 2 step method.

$y' = f(x, y)$

$$\int_{x_n}^{x_{n+1}} y'\, dx = \int_{x_n}^{x_{n+1}} f(x, y(x))\, dx$$

$$y(x_{n+1}) - y(x_n) = \frac{1}{2} h\left[f(x_n, y(x_n)) + f(x_{n+1}, y(x_{n+1}))\right]$$

This is called trapezoidal method. It is an unconditionally stable 2nd order method.

System of eq.

$$y_1' = f_1(x, y_1, \dots y_n)$$
$$y_2' = f_2(x, y_1, \dots y_n)$$
$$\vdots$$
$$y_n' = f_n(x, y_1, \dots, y_n)$$

$f_1, \dots f_n$ are functions with $n+1$ arguments.

Notation

$$Y = (y_1 \dots y_n)^T = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$f = \begin{bmatrix} f_1(x, y_1, \dots, y_n) \\ f_2(x, y_1, \dots y_n) \\ \vdots \\ f_n(x, y_1, \dots, y_n) \end{bmatrix} \qquad \eta = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_n \end{bmatrix}$$

$Y' = f(x, Y)$, $Y(x_0) = \eta$. This is the system IVP.

$y^{(m)} = f(x, y, y', \dots, y^{(m-1)})$

$y = y_1$

$y_1' = y_2$

$\vdots$

$y_{m-1}' = y_m$

$y_m' = f(x, y_1, y_2 \dots y_m)$

$y(a) = \alpha_1$

$y'(a) = \alpha_2$

$\vdots$

$y^{(m-1)}(a) = \alpha_{m-1}$

$\Downarrow$

$y_1(a) = \alpha_1$

$y_2(a) = \alpha_2$

$\vdots$

$y_m(a) = \alpha_{m-1}$

$\therefore$ n^{th} order IVP is equivalent to system of diff. eq. IVP.

Existence & uniqueness of solution

$$y = (y_1, y_2, \dots y_n)^T$$

1. $f(x, y)$ is defined & cont. in the region $a \le x \le b$, $-\infty \le y_i \le \infty$.
2. ∃ a constant L, the Lipschitz constant s.t. for arbitrary $x \in [a, b]$ and any 2 variables y and y^*
$\|f(x, y) - f(x, y^*)\| \le L\|y - y^*\|$

$f(x, y)$ is cont. on B_0 where

$B_0 = \{(x, y) \mid x_0 \le x \le x_0 + a, \; \|y - y_0\| \le b, \; a, b > 0\}$

and satisfies

$\|f(x, y) - f(x, y^*)\| \le L\|y - y^*\| \quad (x, y), (x, y^*) \in B_0$

let $M = \max\limits_{(x, y) \in B_0} \|f(x, y)\|$, $\alpha = \min\left(a, \frac{b}{M}\right)$

The IVP $y' = f(x, y)$, $y(x_0) = y_0$ has a unique solution in $[x_0, x_0 + \alpha]$. This is called Picard - Lindelöf theorem.

Now, system of DE → $y' = \frac{dy}{dx} = f(x, y_1, \dots y_m)$

$y(0) = \eta$

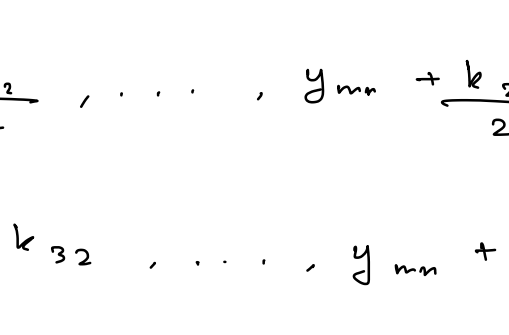
$y = (y_1, \dots y_m)^T$

$\eta = (\eta_1, \dots, \eta_m)^T$

$$f = \begin{bmatrix} f_1(x, y_1, \dots y_m) \\ f_2(x, y_1, \dots y_m) \\ \vdots \\ f_m(x, y_1, \dots y_m) \end{bmatrix}$$

Single step methods

Nodal points



Taylor series method →

$$Y_{n+1} = Y_n + hY_n' + \frac{h^2}{2!}Y_n'' + \dots + \frac{h^p}{p!}Y_n^{(p)}$$

$n = 0, 1, \dots N-1$

$$Y_n^{(k)} = \left[y_{1n}^{(k)}, y_{2n}^{(k)} \dots, y_{mn}^{(k)}\right]^T$$

Classical 4th order RK method

The 4th order RK method becomes →

$$Y_{n+1} = Y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \qquad n = 0, 1, \dots N-1$$

$$k_j = [k_{j1} \; k_{j2} \dots k_{jm}]^T \qquad j = 1, 2, 3, 4$$

$k_{1i} = hf_i(x_n, y_{1n}, \dots y_{mn})$

$k_{2i} = hf_i\left(x_n + \frac{h}{2}, y_{1n} + \frac{k_{11}}{2}, y_{2n} + \frac{k_{12}}{2} \dots, y_{mn} + \frac{k_{1m}}{2}\right)$

$k_{3i} = hf_i\left(x_n + \frac{h}{2}, y_{1n} + \frac{k_{21}}{2}, y_{2n} + \frac{k_{22}}{2}, \dots, y_{mn} + \frac{k_{2m}}{2}\right)$

$k_{4i} = hf_i\left(x_n + h, y_{1n} + k_{31}, y_{2n} + k_{32}, \dots, y_{mn} + k_{3m}\right)$

$i = 1, 2, \dots m$

Euler's method for the system

$\left.\begin{aligned} y' &= f(x, y, z) \\ z' &= g(x, y, z) \end{aligned}\right]$ given the initial conditions $y(x_0) = y_0$, $z(x_0) = z_0$

To find the solution at x_1, x_2 →

$y_{n+1} = y_n + hy_n' = y_n + hf(x_n, y_n, z_n)$

$z_{n+1} = z_n + hz_n' = z_n + hg(x_n, y_n, z_n)$

$n = 0 \Rightarrow y_1 = y_0 + hf(x_0, y_0, z_0)$

$z_1 = z_0 + hg(x_0, y_0, z_0)$

$n = 1 \Rightarrow y_2 = y_1 + hf(x_1, y_1, z_1)$

$z_2 = z_1 + hg(x_1, y_1, z_1)$

$\vdots$

Runge Kutta method of 4th order applied for 2 equations →

To solve $y' = f(x, y, z) \qquad y(x_0) = y_0$

$z' = \phi(x, y, z) \qquad z(x_0) = z_0$

We define $k_1 = hf(x_n, y_n, z_n)$

$l_1 = h\phi(x_n, y_n, z_n)$

$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}, z_n + \frac{l_1}{2}\right)$

$l_2 = h\phi\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}, z_n + \frac{l_1}{2}\right)$

$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}, z_n + \frac{l_2}{2}\right)$

$l_3 = h\phi\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}, z_n + \frac{l_2}{2}\right)$

$k_4 = hf(x_n + h, y_n + k_3, z_n + l_3)$

$l_4 = h\phi(x_n + h, y_n + k_3, z_n + l_3)$

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$z_{n+1} = z_n + \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4)$$

Similarly, we can find y_{n+2}, z_{n+2} by changing n to $n+1$ in the previous step.

2nd order Taylor's method →

$$y_{n+1} = y_n + hy_n' + \frac{h^2}{2!}y_n''$$

$$z_{n+1} = z_n + hz_n' + \frac{h^2}{2!}z_n''$$

Q. Using Euler's method, compute y and z at $x = 1.1, 1.2$ and 1.3, given the following system of ODEs →

$y' = z = f(x, y, z) \qquad y(1) = 0.77$

$z' = -\frac{z}{x} - y = g(x, y, z) \qquad z(1) = -0.44$

Solution

$y_{n+1} = y_n + hf(x_n, y_n, z_n)$

$z_{n+1} = z_n + hg(x_n, y_n, z_n)$

$x_0 = 1, \; y_0 = 0.77, \; z_0 = -0.44, \; h = 0.1$

$y_1 = y_0 + hf(x_0, y_0, z_0)$

$= 0.77 + 0.1 \times (-0.44)$

$\boxed{y_1 = 0.726}$

$z_1 = z_0 + hg(x_0, y_0, z_0)$

$= -0.44 + 0.1 \times (0.44 - 0.77)$

$= -0.44 - 0.033$

$\boxed{z_1 = -0.473}$

Put $n = 1$.

$y_2 = y_1 + hf(x_1, y_1, z_1)$

$z_2 = z_1 + hg(x_1, y_1, z_1)$

$y_2 = 0.726 + 0.1 \times (-0.473)$

$= 0.726 - 0.0473$

$\boxed{y_2 = 0.6787}$

$z_2 = -0.473 + 0.1 \times \left(\frac{0.473}{1.1} - 0.726\right)$

$\boxed{z_2 = -0.5026}$

Solve this for n = 2 now by the same method